RT-FedMTL: A Novel Real-Time Federated Multi-Task Learning Framework with Small Language Models for Natural Language Understanding Tasks

Quang-Vinh Pham and Quang-Hung Le

***Abstract*—Natural language understanding is a crucial component in intelligent systems such as social media monitoring, conversational agents, and information retrieval. Deploying these systems in real application scenarios requires not only accurate but also efficient, privacy-preserving, and capable of handling multiple tasks simultaneously. Centralized training of large language models based approaches raises significant challenges in terms of data privacy, communication overhead, and inference latency. This paper proposes RT-FedMTL, a novel real-time federated multi-task learning framework with small language models for efficient and privacy-preserving natural language understanding. Particularly, our framework unifies federated learning and multi-task learning to enable collaborative training across distributed clients while preserving data locality and leveraging shared representations across tasks. We utilize WebSocket protocols and small language models tuning to reduce the feedback loop and provide near-instantaneous global updates. Extensive experiments conducted on five small language model architectures with respect to three standard natural language understanding tasks to get the best trade-off between performance and resource usage. A significant result demonstrates RT-FedMTL can drastically minimize per-client GPU memory requirements (up to 74-92%) versus centralized multi-task learning baselines, while keeping performance for multiple benchmarks. The source code is available at: https://github.com/vinh1988/RT-FedMTL.git.**

***Index Terms*—Federated Learning, Multi-Task Learning, Federated Multi-Task Learning, Small Language Models, Natural Language Understanding.**

I. Introduction

Natural language understanding (NLU) is a core module in artificial intelligence systems such as social media monitoring [1], conversational agents [2], information retrieval [3], and so on. According to the modern approach, these systems have achieved impressive performance through large-scale language models [4]. However, they typically rely on a centralized training and deployment process which introduces challenges concerning data privacy, computational cost, and inference latency.

In recent years, small language models (SLMs) [5], [6], [7] have emerged as a potential alternative to large models because of that they have low inference latency and cost-effectiveness. Therefore, these models are appropriate in resource-constrained environments, addressing the challenges of large language models [8]. Notwithstanding, SLMs are often limited in capacity, making it hard to maintain performance for a variety of natural language processing (NLP) tasks under independent training.

Parallel to the development of SLMs, multi-task learning (MTL) [9], [10] has become as a powerful paradigm to tackle multiple tasks simultaneously. This approach of leveraging useful information from related tasks can lead to better learning performance [11], [12]. Even so, most existing MTL methods generally require centralized data for training, which is impractical in privacy-sensitive contexts where data is distributed across multiple clients.

At the same time, federated learning (FL) has been applied successfully to domains where data privacy and ownership are critical, such as healthcare [13] and finance [14]. This is a distributed architecture that enables model training across clients without transferring raw data to a central server, thereby preserving privacy while supporting large-scale learning [15], [16]. However, integrating FL with MTL raises significant challenges in terms of communication overhead and inference latency. Furthermore, existing federated NLP frameworks [17], [18], [19] rarely consider real-time constraints, while this is a very important factor for latency-sensitive applications such as health monitoring [20], chatbots [21], and personalized assistants [22]. These challenges motivate the development of real-time federated MTL framework for NLU that reduces communication overhead and inference latency.

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In this work, we propose RT-FedMTL, a novel real-time federated MTL framework with SLMs for efficient and privacy-preserving NLU. Our framework unifies FL and MTL to enable collaborative training across distributed clients while preserving data locality and leveraging shared representations across tasks. Additionally, we utilize WebSocket protocols and SLMs tuning to reduce the feedback loop and provide near-instantaneous global updates. To the best of our knowledge, this is the first work to explore federated MTL with SLMs for real-time NLU. Our key contributions are summarized as follows:

1) We propose a novel real-time federated MTL framework with SLMs for efficient and privacy-preserving NLU. This framework harnesses cross-task knowledge transfer without causing a large amount of resources to be consumed on devices.
2) We assess five SLM architectures (DistilBERT [5], BERT-Medium [6], BERT-Mini [6], MiniLM [7], and TinyBERT [23]) with respect to three standard NLU tasks (sentiment analysis, paraphrase detection, and semantic similarity), to get the best trade-off between performance and resource usage.
3) We investigate the effects of non-independent identically distribution (non-IID) and task heterogeneity on convergence speed and task accuracy in real-time domain.
4) We demonstrate RT-FedMTL can drastically minimize per-client GPU memory requirements (up to 74-93%) versus centralized multi-task learning baselines, while keeping performance for multiple benchmarks.

The remainder of this paper is organized as follows. Section II defines the problem formulation and Section III presents the proposed framework. Section IV describes the experimental setup and results analysis. The related work is briefly presented in Section V. Lastly, Section VI is the conclusion with future work.

II. Problem Formulation

This section introduces the RT-FedMTL environment for NLU. We define the formal mathematical structure for distributed multi-task optimization work, tackle the issues of data and system heterogeneity, and set the global goal for training collaboration models across private edge clients.

A. System Setup

We model a NLU-based RT-FedMTL system in which training is carried out across communication rounds $t = 1, 2, \ldots$ [15], [24], [25]. In each round, a subset of clients is chosen, local updates are made to private data streams, and parameters are returned to a central server for aggregation. As opposed to offline FL, the client data distributions can vary over time. Hence, both local goals and optimum parameters are time-dependent. Let $\mathcal{C} = \{1, \ldots, C\}$ represent the set of clients and $\mathcal{K} = \{1, \ldots, K\}$ represent the set of NLU tasks. Each client $c \in \mathcal{C}$ is assigned to only one main task $\tau(c) \in \mathcal{K}$ (or a small subset of tasks), where new samples are added to the local stream perpetually. The goal is to learn a federated model with cross-task transfer quality but task-level specialization in the absence of IID and time-varying data.

B. Learning Objective

Heterogeneity comes in two forms:

- Heterogeneity of data: the local distributions will differ across clients c_i, c_j (i.e. $P_{c_i}(x, y) \neq P_{c_j}(x, y)$) (label skew, domain shift, vocabulary differences).
- Task heterogeneity: clients might optimize different losses according to $\tau(c)$, which will produce diverging gradients across tasks.

In addition, systems remain heterogeneous, stale updates are propagated by client availability and update frequency. Therefore, the optimization problem must include partial participation and nonuniform local progress. Let $\ell_{\tau(c)}(\theta_e, \theta_{\tau(c)}; \xi)$ be the loss cost for the specific sample $\xi \in \mathcal{D}_c^t$. The universal goal at stage t is

$$\min_{\theta_e, \{\theta_k\}} F_t(\Theta) = \sum_{c=1}^{C} p_c^t \, \mathbb{E}_{\xi \sim \mathcal{D}_c^t} \left[\ell_{\tau(c)}(\theta_e, \theta_{\tau(c)}; \xi) \right] + \lambda \, \Omega(\{\theta_k\}), \tag{1}$$

where p_c^t denote the weight of the aggregated model (i.e., the weight per unit local sample), Ω is a regularizer to enhance the cross-task learning stability and $\lambda \geq 0$ moderates the sharing-specialization trade-off. After local updates, the server also aggregates common encoder parameters through weighted averaging,

$$\theta_e^{t+1} = \sum_{c \in \mathcal{C}_t} \alpha_c^t \, \theta_{e,c}^{t+1}, \tag{2}$$

and updates custom task-specific parameters, following task-wise aggregation,

$$\theta_k^{t+1} = \sum_{c \in \mathcal{C}_t \colon \tau(c) = k} \beta_c^t \, \theta_{k,c}^{t+1}. \tag{3}$$

In $\mathcal{C}_t$ the client subset participates at round t and α_c^t, β_c^t can be viewed as normalized aggregation coefficients [15], [24], [26]. By working towards privacy and heterogeneity requirements, RT-FedMTL is captured as the architecture in real-time, jointly maximizing the optimal use of shared representations and task-specific parameters (2), (3).

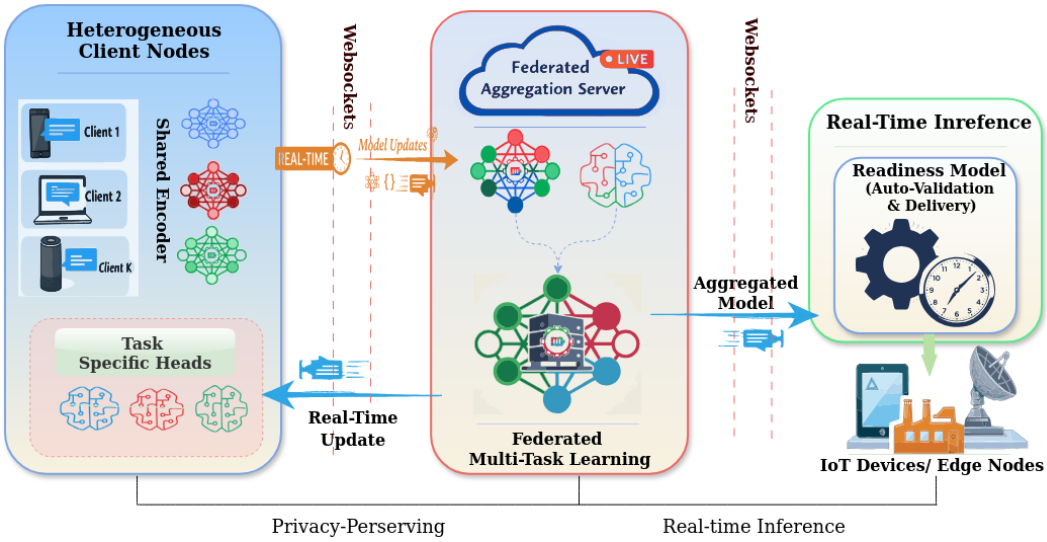

Fig. 1: Architecture of the developed framework for real-time federated MTL.

III. Proposed Framework

The proposed RT-FedMTL simultaneously addresses the challenges of balancing privacy-preserving NLU and resource-constrained edge computing. We introduce a modular federated-based architecture tailored for real-time asynchronous synchronization across WebSockets, multi-task learning (leveraging common linguistic representations with task-specific specialization), and parameter-efficient SLMs to reduce the computational and communication footprint on edge devices.

A. System Overview

As illustrated in Fig. 1, a high level orchestration server, and several heterogeneous client nodes form the basis of the architecture. A parameter server maintains a global state at the center and controls the model aggregation logic. The communications depend on low-latency WebSockets for a bidirectional exchange of model updates, and aggregated parameters.

Federated components and privacy constraints of a system are built on:

- Global server $\mathcal{S}$: orchestrates client sampling, model broadcast, and aggregation.
- Distributed clients $\mathcal{C}$: edge devices/silos to store the raw text.
- Functional modules (f_e, h_k): the shared backbone and task-specific heads parameterized by θ_e, θ_k (see Fig. 1).
- Local datasets $\mathcal{D}_c^t$: private data streams for the client c at round t.

For the sake of privacy, samples are never uploaded. Each of these clients sends only model updates (or compressed adapter and head updates) and acts as host-side aggregation [15], [27]. This is in line with the traditional federated privacy model in which data locality is retained and collaborative training is advocated in the field.

B. Backbone of Small Language Model (SLMs)

This approach enables special SLMs for ensuring an optimal computational efficiency with high language understanding. We use architectures such as DistilBERT [5], BERT-Medium [6], MiniLM [7] for resource-constrained environments. The SLM backbone consists of several integral stages:

1) Tokenizer and Embedding Layer: Outputs are inputted and processed by input text subword tokenizer that converts raw strings into a series of tokens. The generated tokens map to a high-dimensional vector space by means of a learned embedding layer. To model sequence structures without the overhead of recurrences, we add learnable position embeddings at the level of the token embeddings providing the machine with necessary context about the order of the tokens.

2) Low-Cost Transformer Encoder: The central element is a lightweight transformer encoder. These models consist of fewer layers and smaller hidden dimensionality than traditional BERT-base architectures [4]. Each layer utilizes multi-head self-attention for long-range dependencies and the feed-forward network for feature transformation. This streamlined way significantly reduces parameter number and computational load, thus allowing for fast inference and training on edge devices.

3) Shared Representation: The encoder generates a dense vector, represented as a shared representation

$f_e(x;\theta_e)$. This captures the basic semantically and syntactically fundamental information in the input. In our multi-task framework, with the backbone shared across all tasks the model learns universal linguistic patterns while task specific heads focus on unique output. This federated learning-based collaborative tuning guarantees strong generalization across heterogeneous client data.

C. Multi-Task Learning Strategy

We adopt a hybrid approach of sharing the parameters. Throughout the tasks that all share linguistic representation, the lower (encoder) layers of the transformer remains unchanged. On the other hand, top layers are task-centric classification or regression heads respectively. We thus maintain an adaptive loss weighting mechanism to minimize negative transfer:

$$\mathcal{L}_{total} = \sum_{j=1}^{K} w_j \mathcal{L}_j \tag{4}$$

where w_j are task weights dynamically tuned for each task j based on the convergence rate. This formulation ensures that the local update step in Algorithm 1 minimizes the regularized objective function defined in Eq. (1).

We define the parameters

$$\Theta = \{\theta_e, \{\theta_k\}_{k=1}^{K}\}, \tag{5}$$

where θ_e parameterizes the SLM encoder f_e and θ_k parameterizes the task head h_k for task k [25], [26]. Prediction for input x in task k is given by:

$$\hat{y} = h_k\big(f_e(x;\theta_e);\theta_k\big), \tag{6}$$

where $f_e(\cdot)$ and $h_k(\cdot)$ represent the components optimized in Algorithm 1 to maximize sharing while preserving task-specificity.

D. Federated Optimization and Real-Time Adaptation

The federated optimization process is described in Algorithm 1. This system enables real-time adaptation by dynamically synchronizing updates on request via WebSockets. Using a task-aware version of FedAvg [15], the server aggregates updates and normalizes noise from heterogeneous sources (non-IID). Most recent work in adaptive federated optimization [28] suggests incorporating momentum and adaptive learning rates will add an even greater level of stability.

E. Communication-Efficient Fine-Tuning

To further reduce resource usage in edge devices, we perform low-rank adaptation (LoRA) [29]. This ranges from updating in local B, where instead of updating the full weight matrices $W \in \mathbb{R}^{d\times d}$, we optimize the low rank decomposition matrices A and B, where:

$$W' = W + \Delta W = W + BA. \tag{7}$$

$A \in \mathbb{R}^{r\times d}$ and $B \in \mathbb{R}^{d\times r}$ with rank $r \ll d$. When performing big updates, only the matrices A and B are sent, minimizing communication volume up to 99%.

Algorithm 1 RT-FedMTL: Real-time federated MTL

Server Initialize: Shared encoder $\theta_{e,0}$, task heads $\{\theta_{j,0}\}_{j=1}^{K}$
for each round $t = 1, 2, \ldots$ **do**
  Server broadcasts $\theta_{e,t}, \theta_{Task(c),t}$ (for f_e, h_k) to clients $c \in \mathcal{C}_t$
  for each client $c \in \mathcal{C}_t$ in parallel **do**
    Local training: $\{\theta_{e,c}^{t+1}, \theta_{Task(c),c}^{t+1}\} \leftarrow$ LocalUpdate$(\mathcal{D}_c, f_e, h_k, \theta_{e,t}, \theta_{Task(c),t})$ {Optimize $h_k(f_e(\cdot))$}
    Updates $\{\theta_{e,c}^{t+1}, \theta_{Task(c),c}^{t+1}\}$ via WebSockets to Server
  end for
  Encoder Aggregation:
    $\theta_{e,t+1} \leftarrow \sum_{c\in\mathcal{C}_t} \alpha_c^t \theta_{e,c}^{t+1}$ {Aggregate shared backbone f_e}
  for each task $k \in \{1, \ldots, K\}$ in parallel **do**
    $\theta_{k,t+1} \leftarrow \sum_{c\in\mathcal{C}_t:\, \tau(c)=k} \beta_c^t \theta_{k,c}^{t+1}$ {Aggregate specialized head h_k}
  end for
  Server assigns $\{\theta_{e,t+1}, \theta_{Task(c),t+1}\}$ for Local Validation
end for

IV. EXPERIMENTS

In this section, the proposed RT-FedMTL framework is examined in terms of five thematic dimensions: overall baseline comparisons, the impact of MTL, paradigm performance gaps, real-time suitability, and communication efficiency.

A. Experimental Setup

1) Datasets and Tasks: This evaluation combines three core NLU benchmarks that represent different NLU problems from within the GLUE benchmark [11]. SST-2 [30] is a binary classification task (66,477 training, 872 validation, 1,821 test samples) for detecting sentiment polarity. The performance is determined by accuracy and F_1-score [31]. QQP [32] is a binary classification task for paraphrase detection with 323,415 training, 40,431 validation, and 39,081 test samples. Evaluation uses accuracy and F_1-score [31]. STS-B [33] is a regression task that scores the semantic similarity of sentence pairs on a 1 to 5 scale, with 4,249 training, 1,500 validation,

TABLE I: Experimental design matrix

Exp.	Paradigm	Task Config.	Clients	Data Dist.
1	Centralized MTL	Multi-Task	3	Full access
2	Centralized Single	SST-2 only	1	Full access
3	Centralized Single	QQP only	1	Full access
4	Centralized Single	STS-B only	1	Full access
5	Federated MTL	Multi-Task	3	IID
6	Federated MTL	Multi-Task	3	IID + LoRA
7	Federated MTL	Multi-Task	9	Non-IID (3×3)
8	Federated Single	QQP only	3	Non-IID
9	Federated Single	SST-2 only	3	Non-IID
10	Federated Single	STS-B only	3	Non-IID

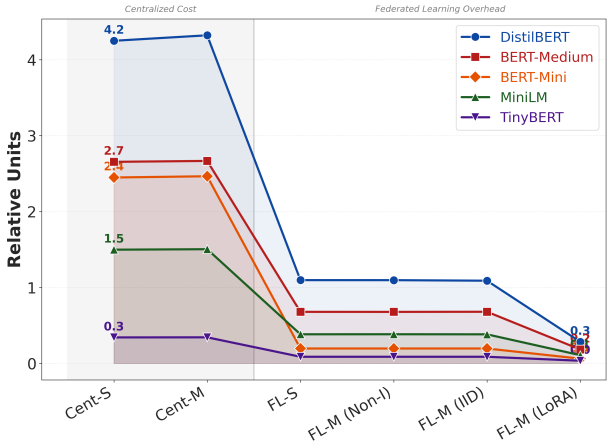

Fig. 2: Resource usage trajectory. Federated settings produce a clear 3-4× reduction in per-client resource usage relative to centralized training because each client holds only a data partition. LoRA further halves resource demands.

and 1,379 test samples. Performance is evaluated by Pearson [34] and Spearman correlation [35].

2) Federated Data Partitioning: The data is uniformly distributed among clients in IID partitioning (Exps. 5-6 in Table I, 3 clients, multi-task scenarios). For non-IID partitioning the Dirichlet distribution (α=0.5), which simulates real heterogeneity of data, uses three clients per task (9 total), giving 107,805 / 22,159 / 1,416 samples per client for QQP, SST-2, and STS-B respectively (Exps. 7-10 in Table I). Validation sets are *not* split over clients in any paradigm, which guarantees global model assessment consistency.

3) Baselines: Three baselines are tested: (1) centralized MTL: all 5 SLM types trained on centralized data and share encoder and task head, this is used as the top-tier reference (Exp. 1 in Table I); (2) federated single-task: per-task federated experiments with 3 clients and non-IID data (Exps. 8-10 in Table I); and (3) large-model federated methods: evaluation against larger language models to see communication efficiency and performance trade-offs.

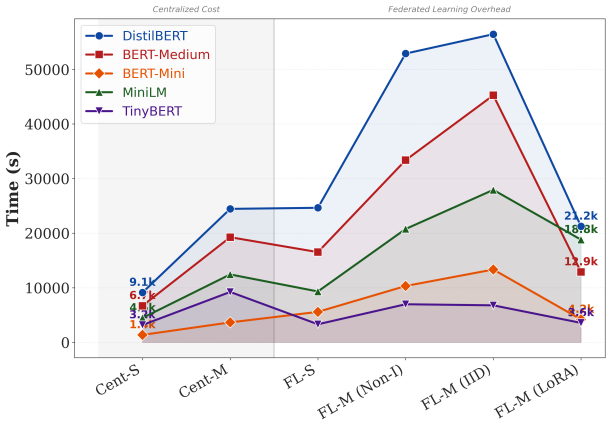

Fig. 3: Training time trajectory capturing the wall-clock training durations. The transition from centralized to federated multi-task settings introduces a significant overhead, which LoRA substantially offsets by limiting parameter transmission.

4) Evaluation Metrics: Task-level metrics are accuracy & F_1-score [31] (SST-2, QQP) and Pearson [34] / Spearman correlation [35] (STS-B) [11]. Federated system-level metrics are communication cost per round, convergence speed, and total training time.

5) Implementation Details: All experiments use learning rate 2×10^{-5} with 1,000-step warmup, batch size 8 for all experiments (centralized to FL/FL-MTL), weight decay 0.01, gradient clipping at the max-norm of 1.0, and random seed 42. Centralized are trained with 25 epochs; federated clients train 1 local epoch per round. Communication is provided by ports 8771-8775 with 60 s client timeout and 3,400 s round timeout and 3-retry exponential-backoff.

B. Results and Discussion

1) Overall Performance (Comparison across Baselines): Table IV shows that SLMs within RT-FedMTL still achieve good performance in NLU. Centralized MTL is at least a baseline; however, DistilBERT in federated MTL (IID) performance, surprisingly, is 0.9461 on SST-2, better than that of its centralized counterpart. MiniLM features a very low resource-to-function ratio with 0.8966 and 0.8384 in QQP and STS-B tasks using IID respectively-scores that are on par or better than those of much larger models and taking only 0.38 GB of GPU memory (compared with 1.09 GB in DistilBERT). It confirms that in distributed NLU the size of the model does not only matter for the quality, and compact architectures such as MiniLM can be fine-tuned with respect to it.

2) Effect of Multi-Task Learning: MTL offers notable benefits in federated scenarios on account of its

TABLE II: Resource usage trajectory (GB, per-client average)

Model	Cent-S	Cent-M	FL-S	FL-M (Non-IID)	FL-M (IID)	FL-M (LoRA)
DistilBERT	4.2492	4.3214	1.0972	1.0968	1.0896	0.2852
BERT-Medium	2.6549	2.6671	0.6795	0.6790	0.6807	0.1840
BERT-Mini	2.4487	2.4651	0.1969	0.1969	0.1969	0.0630
MiniLM	1.4982	1.5045	0.3839	0.3846	0.3835	0.1090
TinyBERT	0.3427	0.3440	0.0881	0.0881	0.0881	0.0356

Cent-S: Centralized Single-Task, Cent-M: Centralized Multi-Task, FL-S: Federated Single-Task, FL-M: Federated Multi-Task, IID/Non-IID: Data Distribution.

TABLE III: Training time trajectory (seconds)

Model	Cent-S	Cent-M	FL-S	FL-M (Non-IID)	FL-M (IID)	FL-M (LoRA)
DistilBERT	9,122	24,450	24,648	52,904	56,454	21,245
BERT-Medium	6,693	19,257	16,527	33,375	45,258	12,886
BERT-Mini	1,359	3,653	5,581	10,318	13,341	4,192
MiniLM	4,473	12,437	9,317	20,766	27,930	18,808
TinyBERT	3,170	9,228	3,308	6,975	6,778	3,561

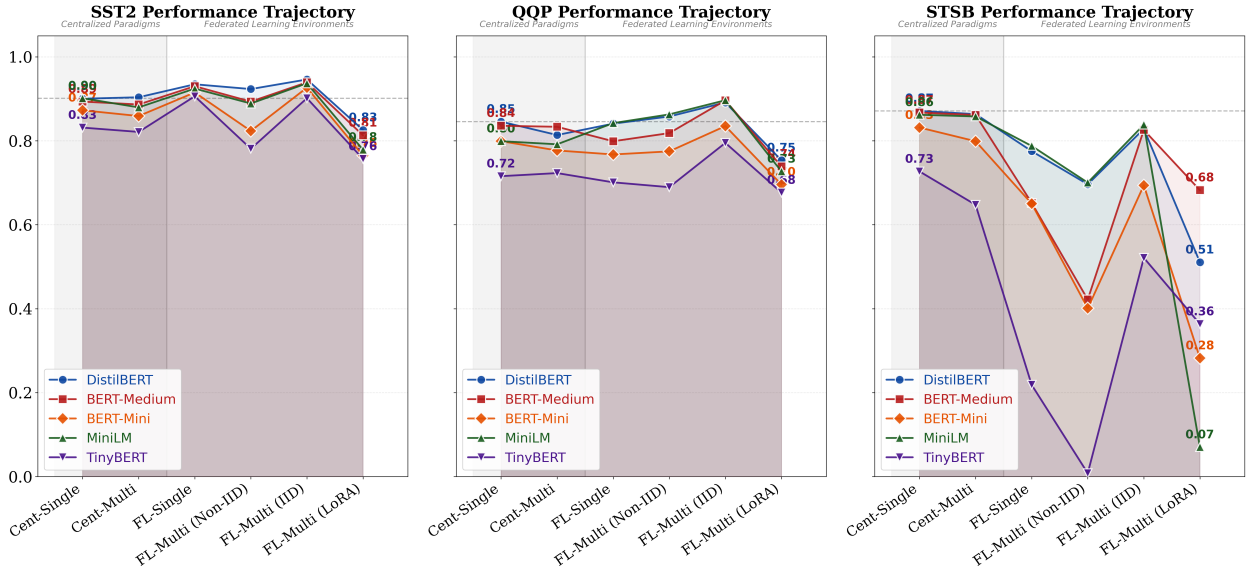

Fig. 4: Performance across six paradigm-complexity stages for five SLM architectures. Dashed lines separate centralized and federated settings.

implementation of shared representations across similar tasks. As evidenced in Figure 4 and Table IV, it is evident that comparing FL-single and FL-MTL (non-IID) has rich task interplay. SST-2 shows mild negative transfer in some setups, whereas it is usually worse for regression problems such as STS-B for non-IID situations. Nevertheless, classification tasks tend to demonstrate more positive transfer, with features learned by SST-2 benefiting QQP for example. Large SLMs such as DistilBERT and BERT-Medium exhibit robustness to the negative transfer, because they have a higher representational ability.

3) Federated Learning Performance: A "reliability gap" emerges as the switching from centralized to federated learning is observed from a moderate quality ($\leq 5\%$) degradation in classification for tasks such as SST-2 and QQP. This difference is most prominent in the STS-B Pearson correlation, which decreases notably for smallest architectures such as TinyBERT under the non-IID MTL scenario. This line is illustrated in Figure 4, which illustrates how a proposed RT-FedMTL does in fact mitigate this gap by stabilizing performance without loss of signal across tasks. Notably, DistilBERT is generally the most stable architecture for sentiment-heavy tasks, as it shows only a slow decay performance even under the worst of non-IID skew (0.9232 Acc. on SST-2). The performance of IID federated results, in a quantitative sense, confirms that MTL can generate better results

TABLE IV: Comparative analysis of SLM performance across paradigms (— denotes not applicable)

Model	Paradigm	Task	Dist.	Tuning	SST-2		QQP		STS-B		System	System
					Acc.	F_1	Acc.	F_1	Pear.	Spea.	Time (s)	Res (GB)
DistilBERT [5]	FL	ST	Non-IID	FFT	—	—	0.8408	0.7724	—	—	34904.44	1.0956
	FL	ST	Non-IID	FFT	—	—	—	—	0.7753	0.7421	15452.52	1.0988
	FL	ST	Non-IID	FFT	0.9346	0.9392	—	—	—	—	23586.44	1.0972
	FL	MTL (ours)	IID	LoRA	0.8257	0.8433	0.7535	0.6754	0.5108	0.4919	21245.26	0.2852
	FL	MTL (ours)	Non-IID	FFT	0.9232	0.9298	0.8579	0.8054	0.6963	0.6967	52903.97	1.0968
	FL	MTL (ours)	IID	FFT	0.9461	0.9514	0.8913	0.8610	0.8284	0.8136	56454.46	1.0896
	Cent.	ST	IID	FFT	—	—	—	—	0.8712	0.8691	4561.78	4.2470
	Cent.	ST	IID	FFT	0.9002	0.9061	—	—	—	—	18514.82	4.2460
	Cent.	ST	IID	FFT	—	—	0.8456	0.8934	—	—	4290.59	4.2547
	Cent.	MTL	IID	FFT	0.9037	0.9077	0.8137	0.8778	0.8635	0.8614	24449.68	4.3214
BERT-Medium [6]	FL	ST	Non-IID	FFT	—	—	0.7989	0.6852	—	—	23880.89	0.6789
	FL	ST	Non-IID	FFT	—	—	—	—	0.6521	0.6212	10261.92	0.6810
	FL	ST	Non-IID	FFT	0.9300	0.9348	—	—	—	—	15439.04	0.6786
	FL	MTL (ours)	Non-IID	FFT	0.8933	0.8988	0.8183	0.7441	0.4219	0.4132	33374.50	0.6790
	FL	MTL (ours)	IID	FFT	0.9392	0.9450	0.8962	0.8655	0.8258	0.8052	45258.21	0.6807
	FL	MTL (ours)	IID	LoRA	0.8131	0.8362	0.7388	0.6805	0.6823	0.6512	12885.74	0.1840
	Cent.	ST	IID	FFT	—	—	—	—	0.8673	0.8648	3381.83	2.6467
	Cent.	ST	IID	FFT	0.8933	0.8970	—	—	—	—	13492.62	2.6711
	Cent.	ST	IID	FFT	—	—	0.8358	0.8859	—	—	3205.51	2.6467
	Cent.	MTL	IID	FFT	0.8865	0.8884	0.8333	0.8777	0.8615	0.8584	19257.40	2.6671
BERT-Mini [6]	FL	ST	Non-IID	FFT	—	—	0.7674	0.6201	—	—	9642.59	0.1969
	FL	ST	Non-IID	FFT	—	—	—	—	0.6504	0.6150	2547.10	0.1969
	FL	ST	Non-IID	FFT	0.9151	0.9196	—	—	—	—	4552.75	0.1969
	FL	MTL (ours)	Non-IID	FFT	0.8234	0.8226	0.7748	0.6668	0.4009	0.3939	10317.55	0.1969
	FL	MTL (ours)	IID	LoRA	0.7649	0.7915	0.6958	0.5602	0.2824	0.2773	4191.67	0.0630
	FL	MTL (ours)	IID	FFT	0.9266	0.9329	0.8350	0.7981	0.6940	0.6550	13341.01	0.1969
	Cent.	ST	IID	FFT	—	—	—	—	0.8316	0.8283	411.45	2.4487
	Cent.	ST	IID	FFT	0.8727	0.8773	—	—	—	—	3348.26	2.4487
	Cent.	ST	IID	FFT	—	—	0.7990	0.8536	—	—	318.61	2.4487
	Cent.	MTL	IID	FFT	0.8589	0.8650	0.7770	0.8580	0.7989	0.7973	3652.50	2.4651
MiniLM [7]	FL	ST	Non-IID	FFT	—	—	0.8421	0.7644	—	—	14583.05	0.3842
	FL	ST	Non-IID	FFT	—	—	—	—	0.7877	0.7489	4920.48	0.3840
	FL	ST	Non-IID	FFT	0.9243	0.9293	—	—	—	—	8447.20	0.3836
	FL	MTL (ours)	IID	FFT	0.9369	0.9428	0.8966	0.8662	0.8384	0.8197	27930.06	0.3835
	FL	MTL (ours)	Non-IID	FFT	0.8888	0.8974	0.8629	0.8150	0.7003	0.7061	20765.57	0.3846
	FL	MTL (ours)	IID	LoRA	0.7787	0.8102	0.7271	0.6060	0.0703	0.0621	18808.08	0.1090
	Cent.	ST	IID	FFT	—	—	—	—	0.8620	0.8592	2696.27	1.4929
	Cent.	ST	IID	FFT	0.9014	0.9051	—	—	—	—	8334.10	1.4950
	Cent.	ST	IID	FFT	—	—	0.7990	0.8656	—	—	2388.13	1.5067
	Cent.	MTL	IID	FFT	0.8796	0.8829	0.7917	0.8649	0.8580	0.8560	12437.12	1.5045
TinyBERT [23]	FL	ST	Non-IID	FFT	—	—	0.7010	0.3887	—	—	6226.63	0.0881
	FL	ST	Non-IID	FFT	—	—	—	—	0.2186	0.1755	1182.03	0.0881
	FL	ST	Non-IID	FFT	0.9060	0.9122	—	—	—	—	2514.06	0.0881
	FL	MTL (ours)	IID	FFT	0.9014	0.9145	0.7950	0.7494	0.5212	0.4754	6778.00	0.0881
	FL	MTL (ours)	Non-IID	FFT	0.7810	0.7782	0.6895	0.4733	0.0081	0.0063	6975.28	0.0881
	FL	MTL (ours)	IID	LoRA	0.7569	0.7902	0.6766	0.4475	0.3636	0.3505	3560.68	0.0356
	Cent.	ST	IID	FFT	—	—	—	—	0.7270	0.7230	2337.42	0.3427
	Cent.	ST	IID	FFT	0.8314	0.8358	—	—	—	—	4857.08	0.3427
	Cent.	ST	IID	FFT	—	—	0.7157	0.8139	—	—	2315.39	0.3427
	Cent.	MTL	IID	FFT	0.8211	0.8263	0.7230	0.8101	0.6474	0.6426	9227.55	0.3440

ST: Single-Task, MTL: Multi-Task Learning, Cent.: Centralized, FFT: Full Fine-Tuning, Pear.: Pearson Correlation, Spea.: Spearman Correlation, —: Not applicable.

over an isolated centralized training for discrete model performance (e.g., DistilBERT reaching 0.9461 on SST-2). The non-IID metrics highlight that the framework is more robust to data heterogeneity; classification tasks such as SST-2 retain steady performance (e.g., 0.9232 for DistilBERT), but regression tasks such as STS-B show a greater degree of sensitivity. MiniLM is consistently the greatest versatile candidate in both IID and non-IID distributions being able to win 2 of 3 benchmark tasks (QQP and STS-B) with a memory footprint savings of nearly 3× when compared to DistilBERT. This shows that although the framework is robust to categorical benchmarks, the selection of the model needs to be a delicate bit better, where DistilBERT for sentiment robustness and MiniLM for resource-constrained multi-task stability.

4) Real-Time Inference Analysis: For deployment on edge devices where real-time inference is critical, resource consumption and model footprint are the defining metrics. Figure 2 and Table II highlight that FL reduces individual client GPU memory demands by 74-92% relative to centralized training, as calculated by:

$$\text{Reduction} = \frac{M_{\text{Cent-M}} - M_{\text{FL-M}}}{M_{\text{Cent-M}}} \times 100\% \tag{8}$$

where $M_{\text{Cent-M}}$ and $M_{\text{FL-M}}$ denote the average per-client GPU memory usage for the centralized multi-task learning and federated multi-task learning (IID) settings respectively, as detailed in Table II. TinyBERT and BERT-Mini emerge as the leading candidates for extremely resource-constrained devices ($< 0.1\,\text{GB}$ and $< 0.2\,\text{GB}$ per client respectively), albeit with bounded performance on complex tasks. MiniLM and DistilBERT provide a superior performance-efficiency balance for most real-time scenarios, maintaining high accuracy while keeping per-client resource usage significantly lower than the centralized baseline.

5) Communication Efficiency: Communication overhead is the primary bottleneck in federated MTL, with wall-clock training times increasing by 2.2-3.7× compared to centralized training (Table III). However, as illustrated in Figure 3, the integration of parameter-efficient fine-tuning (PEFT) via LoRA [29] substantially mitigates this cost. LoRA reduces federated training time by 38-62% for most models by transmitting only low-rank adapter weights instead of full model parameters. This optimization enables the framework to maintain competitive accuracy while drastically reducing the bandwidth requirements, making it suitable for bandwidth-limited edge deployments.

V. Related Work

SLMs including DistilBERT [5], MiniLM [7], and ALBERT [36] demonstrate that strict compression and distillation can maintain fair performance at a cost of parameter number in the balance with inference. The common reduction methods are knowledge distillation, parameter sharing, low-rank factorization, and architectural simplification. These resources make SLMs particularly well suited to edge and distributed learning: transmission overhead is minimized, on-device learning cost on-device is significantly reduced and federated iteration cycles are speeded up. Recent advancements such as EdgeDistilBERT [37] further improve on the state-of-the-art lightweight multi-task FL on edge devices. In federated multi-task systems, SLMs also allow scaling to larger populations of clients and tasks on resource budgets for the system.

MTL for NLP usually uses one of the two paradigms: hard parameter sharing or soft parameter sharing [24], [12]. Unlike many of the popular methods such as adversarial sharing which assumes a task head is separate from each model instance, hard sharing utilizes a common encoder for every model instance, which is useful to save the model from overfitting and cost of training. Meanwhile, soft sharing retains task-specific factors, but promotes similarity in some way through regularization and/or through low interaction. Also, PEFT, with adapters [38] and modular shared frameworks such as AdapterHub [39], has allowed rapid multi-task adaptation with less overhead. MTL enhances representation learning by transferring useful linguistic data from a task to another. But this also introduces known risks: task imbalance (over-representation of dominant tasks with lower-level tasks) and negative transfer (one task injuring others). These difficulties are especially high in the federated training case, since the client participants involved in the training process are partially responsible, and the balance of distribution of the data over the rounds is not uniform.

The non-IID data remains one of the worst bottleneck in the FL process. This can lead to biased updates at local level and cause global optimization failures at global level due to client-dependent language patterns, domain shifts, and label bias. In heterogeneous NLU deployments, existing differences of client tasks (cross-client task diversity) leads to partial representations of client to serve multiple objectives, exacerbating this issue. To mitigate these heterogeneity effects, strategies such as FedBN [40] and low-rank parameterizations like FedPara [41] have been proposed. As in past work, aggregation without considering the structure of a task will reduce velocity and final generalization, and mixing heterogeneity (to aggregate) will degrade speed. Due to the fact convergence velocity may hinder good generalization and final speed depending on task structure, there have been some proposed techniques, like personalized layers, proximal regularization, gradient conflict reducing and encoder–decoder decomposition which bridged this gap [42], [12], [26].

FL in NLP/NLU applications include text classification and personalized language services where raw text cannot be aggregated [15], [42]. Fundamental optimization algorithms such as FedAvg [15] which defines the client–update–server aggregation loop after completion, and FedProx [42] performed regularization to enable enhanced behavior to counter client drift. Further improvements, such as SCAFFOLD [43], mitigate gradient heterogeneity to guarantee convergence stability. Federated MTL methods like MOCHA illustrate that inter-task relatedness can be strengthened under heterogeneous settings through enhanced statistical efficiency and stability to system issues [24]. Newer approaches such as FedBone [25], MIRA [27] and M-Fed [26] take this the step further by separating between shared and task-dependent elements and even specialized encoder–decoder architectures and adapting these into large scale heterogeneous MTL environments. However, several FL for NLP studies are still based on very little task diversity or the individual personalization of user at single task.

VI. CONCLUSION

In this work, we introduced RT-FedMTL, a novel real-time federated MTL framework with SLMs for efficient and privacy-preserving NLU. Extensive experiments conducted on five SLM architectures with respect to three standard NLU tasks to get the best trade-off between performance and resource usage. A significant result shows that our framework can drastically minimize per-client GPU memory requirements by 74-92% versus centralized MTL baselines, while keeping performance for multiple benchmarks. Furthermore, the integration of PEFT via LoRA reduces federated training time by 38-62% for most models by transmitting only low-rank adapter weights instead of full model parameters. This empirical evidence demonstrates that LoRA can handle communication bottlenecks in real-time scenario remarkably well.

As a follow-up in the future, we plan to investigate adaptive task-sharing mechanisms to avoid negative transfer. In addition, we further explore personalized federated learning techniques in the real-time communication layer, allowing the efficiency gains not to come at the cost of the privacy in the production NLU deployments.

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